

AYMANE AIT DADS

Data / ML / AI Intern | Machine Learning · Statistical Analysis · Python Development

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PROFESSIONAL SUMMARY

Ranked 17th out of 3,000+ teams (score 0.86/1.0, \$106K+ prize pool) in the NVIDIA Nemotron Reasoning Challenge by fine-tuning a 30B-parameter model with LoRA SFT, demonstrating deep expertise in machine learning and statistical analysis methods. Built Random Forest and K-Means clustering pipelines on 100K+ telecom records, reducing manual analysis time by 60% and mapping customer profiles to 5 commercial service tiers used directly by an optimization team. Brings strong Python development skills, ensemble modeling, rigorous ablation methodology, and end-to-end data science experience spanning NLP, computer vision, and statistical modeling. Aims to apply quantitative ML and data analysis skills at Dialectic to extract signal from complex on-chain and financial market datasets, supporting research and product development in the DeFi space.

CORE COMPETENCIES

Machine Learning Methods | Statistical Analysis | Python Development | Financial Data Analysis | Database Software | Entrepreneurial Mindset | Data Science | Clustering & Ensemble Methods

WORK EXPERIENCE

Orange Maroc

Jul 2024 - Aug 2024

Data Science Intern - Casablanca, Morocco

- Built **Random Forest and K-Means machine learning models** on 100K+ fiber-optic network records (upload speed, latency, jitter) to classify network quality across thousands of nodes.
- Developed a **clustering pipeline** mapping customer performance profiles to 5 commercial service tiers, with output adopted directly by the network optimization team for production decisions.
- Reduced manual analysis time by **~60%** through automated feature engineering, statistical analysis, and systematic model evaluation using Python, Scikit-learn, and Pandas.
- Delivered stakeholder presentations translating **model outputs and data insights** into actionable maintenance recommendations, bridging technical findings and business decision-making.

PROJECTS

NVIDIA Nemotron Model Reasoning Challenge

Kaggle Competition · In Progress (June 2026) · Ranked 17th / 3,000+ Teams

- Ranked **17th / 3,000+ teams** (score 0.86/1.0, \$106K+ prize pool) in the NVIDIA Nemotron Reasoning Challenge by fine-tuning a **30B-parameter Mamba-Transformer** with LoRA SFT (r=32, BF16, completion-only loss, 8,192-token context).
- Performed **task-level error analysis** across 5 reasoning categories (unit conversion, cipher, symbolic, gravity, bit manipulation); built a diagnostic evaluation pipeline revealing that over-expanded reasoning - not incorrect reasoning - was the primary failure mode across hard tasks.

PyTorch · Unsloth · TRL · PEFT · Hugging Face Transformers · vLLM · Kaggle GPU

AI-Generated Image Detection

EURECOM ImSecu + NTIRE 2026 @ CVPR · 2025 · Ranked 1st in Class

- Ranked **1st in the EURECOM class private Kaggle leaderboard** (0.791 AUC) by fine-tuning CLIP ViT-L/14 (428M params) with a custom MLP head, dual learning rates, and FP16 mixed precision training on 250K images across 25+ generator types.
- Submitted to the **NTIRE 2026 @ CVPR CodaBench** international benchmark; applied 10-view TTA (flips, crops, blur, resizes) and a 3-model ensemble weighted by validation AUC, with a key finding that 1-epoch training (0.793 AUC) outperformed 4-epoch training (0.767 AUC) for out-of-distribution generalization.

PyTorch · OpenCLIP · ViT · AdamW · Kaggle GPU

EDUCATION

Engineering Degree in Data Science (Master-Level) - EURECOM

Sep 2023 - Present

Relevant coursework: Machine Learning, Deep Learning, Statistics, Cloud Computing, Computer Vision, NLP, Image Security, Web Applications

CPGE - Mathematics & Physics - Ibn Timiya

2021 - 2023

SKILLS

Machine Learning & Statistical Analysis: Python | Scikit-learn | PyTorch | TensorFlow | Pandas | NumPy | Ensemble Methods | Clustering

LLM Fine-Tuning & NLP: LoRA | SFTTrainer | PEFT | Hugging Face Transformers | vLLM | Chain-of-Thought | Prompt Engineering

Data Engineering & Infrastructure: SQL | PostgreSQL | Power BI | Git | Linux | Docker | REST APIs | Feature Engineering